Final Project: Open Research in Conversational AI

ECE 6256 / CS 6604 · Fall 2026 · @Larry Heck · Oct 26, 2026

Overview

Your team of four will pick an unsolved problem in conversational AI, build a system that attacks it, test it with experiments, and write it up as a paper of at least workshop quality. The project is **30%** of the course grade, and the final paper and code are due Dec 8, 2026 at 11:59 pm.

It is the case-study workflow turned toward original research. There are three differences from the case studies:

1. **Your team picks the topic,** and it must be a current research problem: open, not solved.
2. **You build and test something.** An initial system is coded and evaluated through experiments.
3. **You write it up as a research paper.** Its Introduction and Background read like your case-study surveys; the rest reports your own work.

Teams are the same teams of 4 as the case studies. *Workshop quality* means a reviewer at a venue such as SIGDIAL, NLP4ConvAI, or an ACL or Interspeech workshop would rate the paper borderline accept or better. Strong projects are encouraged to submit for real.

What counts as research

A project qualifies if it asks a question nobody has answered yet, and a negative result would still teach the field something. Your proposal must pass all five tests.

Test	What it means	It fails if…
Open	The problem is unsolved: you cite the closest prior work and state the gap	A paper already answers it, or no prior work is cited

Test	What it means	It fails if…
Answerable	A hypothesis or research question that an experiment can support or refute	The goal is "build X" with no question
Anchored	Rooted in one or more course lectures, named in the proposal	The topic sits outside conversational AI
Measured	A baseline that is not a straw man, metrics that fit the question, multi-turn evaluation when the claim is about conversation	The evidence is anecdotes or a single demo
Feasible	Fits about five weeks and the compute your team can actually get	It needs frontier-scale training, or data you do not have yet

These do not qualify on their own: a product or app with no research question; a reproduction with no extension (a good first step, not a contribution); a survey (that was the case studies); a prompt-engineering demo without controlled comparisons.

AI tools

The project follows ACL's rules on generative AI, exactly as if you were submitting to an ACL venue: the ACL Policy on Publication Ethics, §4.2.2, Guidelines for Generative Assistance in Authorship, which ACL Rolling Review applies to every submission.

- **AI tools cannot be authors.** Every team member is accountable for all of the paper's content: text, code, numbers, and citations.
- **Only the uses ACL lists are allowed:** polishing your own text, short-form input assistance, literature search, low-novelty text that you check and cite, and model-suggested ideas that you disclose. ACL treats any other use as inappropriate, and a model that supplies both new ideas and the text presenting them would be doing an author's job.
- **Disclose what you used** in an Acknowledgements section: which tools, for what, and how you checked the results. For coding assistance, say so in your README too. Proofreading tools such as spell checkers need no disclosure. This section is your course AI-Use Disclosure.

- **Every citation must be real and read.** References found with AI tools carry the usual accuracy requirements. A fabricated or misattributed citation is an integrity issue, not a typo.
- **No hidden prompts** or other content meant to steer automated reviewing.

One more item goes in an appendix:

- **Team contribution statement.** One line per member on what they did; CRediT-style roles are fine.

Choosing a problem

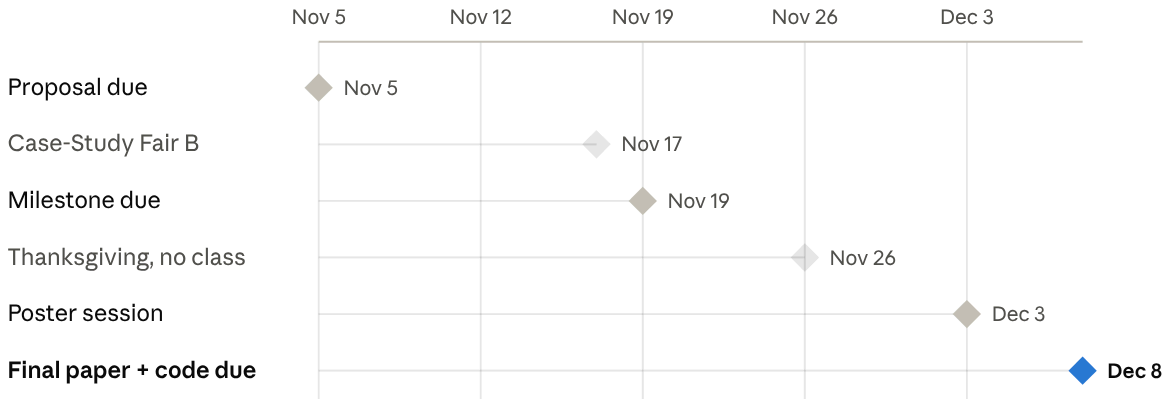
Start from a lecture, not a technology: pick the part of conversational AI you want to move, then find what is still open there. The seeds below show the right size of problem. They are examples, not a menu, and your own idea is welcome if it passes the five tests.

Unit (lectures)	Example open problems
1 · Classical decomposition (L1–L3)	Can an LLM agent keep a calibrated belief state over slots across a long dialogue, and does exposing that state improve repair?
2–3 · Neural foundations and LLMs (L4–L11)	Which post-training choices produce the generic assistant persona in multi-turn dialogue, and can they be undone cheaply? Latency-aware decoding for conversational turns
4 · Evaluation and user simulation (L12)	Closing the gap between simulated and real users; detecting goal drift in LLM user simulators; LLM-as-judge bias on multi-turn transcripts
5 · Knowledge, memory, action (L13–L17)	Conversational query rewriting when dense retrieval is mostly lexical overlap; memory that survives multi-turn drift; cross-lingual unlearning; emotional framing and reasoning; reward design for multi-turn tool use
6 · Voice (L18–L21)	Disaggregated ASR error against task success; backchannel and barge-in timing in full-duplex models; how much task ability voice agents lose against text

Unit (lectures)	Example open problems
7 · Multi-agent–human and safety (L22–L23)	When an agent should speak up in a multi-party chat; whether latent (KV-cache) messages between agents actually carry information; persuasion and sycophancy cascades among agents; dialogic deference
8 · Latent structure (L24)	Probing for interaction moves rather than intents; steering one move without breaking content; small controllers over frozen speech LLMs
Multimodal (optional readings)	Outside-knowledge conversational video QA; multimodal form filling; situated reference resolution

Deliverables and dates

There are four deliverables, each building on the one before. One member submits for the team on Gradescope.



project timeline · 4 deadlines, Nov 5 to Dec 8

The milestone lands two days after Fair B, so Fair B teams should have their baseline running before fair prep peaks.

Proposal

Due Nov 5, 2026, 11:59 pm · 2 pages in ACL format, plus references.

- Title, team, and the lecture(s) the project is anchored in
- The problem and why it is open: the closest prior work (at least 5 citations) and the gap
- Hypothesis or research questions
- Method, and the system you will build
- Data, baselines, metrics, and how you will evaluate multi-turn behavior
- Compute plan: what you need and where it comes from
- Risks, and a fallback that still yields a paper
- Division of labor

You will get feedback within a week. Some teams may be asked to re-scope.

Milestone

Due Nov 19, 2026, 11:59 pm · 4 pages in ACL format, plus references.

- Draft Introduction and Background
- A running baseline, with at least one real result
- What changed since the proposal, and the plan for the last three weeks

Poster

Presented in class Dec 3, 2026 · poster PDF due the night before, 11:59 pm.

- Two shifts, as at the case-study fairs: each team presents during one shift and reviews during the other
- Each student submits a short structured critique of 4 assigned posters (credit or no credit, counted in participation)
- Use the feedback: the final paper is due five days later

Final paper and code

Due Dec 8, 2026, 11:59 pm · up to 8 pages in ACL format, not counting references, Limitations, Ethics, and appendices.

- An Acknowledgements section disclosing AI use (see AI tools), plus an appendix with the team contribution statement
- A link to your code repository (see the next section)
- Dec 8 is the last instructional day, so this deadline cannot move

Requirements for the final paper and code

A paper that meets every item below can earn full marks; a missing item caps the matching rubric line.

Paper

- Sections: Abstract, Introduction, Background, Method, Experiments, Results, Analysis, Limitations, Ethics, Conclusion, Acknowledgements
- At least one baseline you did not design to lose, and at least one ablation of your own method
- Results with variability: confidence intervals, standard deviations over seeds, or significance tests, not single runs
- Multi-turn evaluation whenever the claim is about conversation; per-turn scores alone do not show a dialogue went well
- Error analysis with real examples, including where your method fails
- Disaggregated results where the population matters (accents, languages, user types)

Code and data (repository link in the paper: GitHub, Georgia Tech GitHub Enterprise, or a zip on Canvas)

- A README with setup steps and one command or notebook per main table or figure
- Fixed seeds, pinned dependencies, and every prompt you used, verbatim
- A data statement: sources, licenses, any personal data, and how you handled it
- Compute used: hardware, plus approximate GPU hours or API spend

Appendix: the team contribution statement (see AI tools).

Grading

The project is 30% of the course grade, and the final paper carries 18 of those points.

Deliverable	Share of course grade	Graded on
Proposal	3%	Passes the five tests; a clear, feasible plan
Milestone	4%	A running baseline, real results, a credible plan
Poster	5%	Clarity, correctness, and handling questions

Deliverable	Share of course grade	Graded on
Final paper and code	18%	The rubric below
Total	**30%**	

The final paper is scored out of 100:

Criterion	Points	A full-marks paper…
Problem and novelty	20	Poses an open, well-motivated question and states the gap against accurate prior work
Method and experiments	25	Tests the question with fair baselines, ablations, and fitting (multi-turn) metrics
Results and analysis	20	Reports variability and explains why things worked or failed, with examples
Scholarship and writing	15	Reads like a workshop paper; every citation is real and correctly attributed
Reproducibility	10	Lets someone rerun the main results from the repository
Responsible AI use	10	Follows the ACL policy and discloses AI use completely and candidly

Teammates get the same grade by default. A confidential peer evaluation, submitted with the final paper, can lower an individual's project grade when contributions were clearly unequal.

Policies

- **Slip days.** Slip days apply to the proposal and the milestone, at most two per deliverable; a slip day on a team deliverable is charged to every member. The poster session and the Dec 8 final paper cannot be extended.
- **AI use.** Follow the ACL policy (see AI tools). Humans are accountable for every claim, number, and citation. A candid disclosure never costs points on the work itself; an empty or evasive one on work that plainly used AI is an Honor Code matter.

- **Academic integrity.** Fabricated results or citations, and undisclosed reuse of others' work, violate the Georgia Tech Honor Code. Building on your own or your lab's earlier work is fine if you disclose it and the project adds a distinct contribution.
- **Human subjects.** A user study, or any data from people outside your team, needs my approval before the proposal is due. Work you intend to publish usually needs Georgia Tech IRB review, which can take longer than the project, so plan a fallback on simulated or existing data.
- **Compute.** Size the project to the compute your team can actually get, and say in the proposal where it comes from. Talk to me early if your plan needs GPUs you do not have.
- **Submitting for real.** Strong projects are encouraged to go to a workshop. Because the project already follows ACL's policy, an ACL submission mainly adds the Responsible NLP checklist's question on AI assistants.

FAQ

Can we continue our case-study topic? Yes, if you move from surveying it to an open question and an experiment.

Can we use our lab's research? Yes, with disclosure (see Academic integrity). The course project must be a distinct contribution, and all four members must contribute to it.

What if our main result is negative? A well-run negative result with a clear explanation can earn full marks. An experiment that never ran cannot.

Can we change topics after the proposal? Up to the milestone, with my approval. After that, narrow the question rather than switch it.

Do we need to beat the state of the art? No. You need a question that matters, a fair test, and an honest account of what you found.